

Laplace Transform

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

06016401

MATHEMATICS FOR INFORMATION TECHNOLOGY

Instructor: Asst. Prof. Dr. Praphan Pavarangkoon

Office: 4th Floor

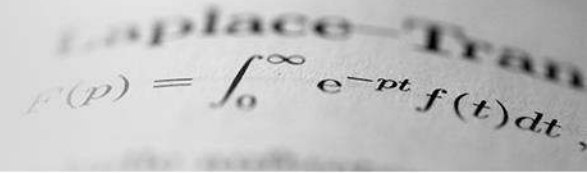
Email: praphan@it.kmitl.ac.th

Office hours: Thursday at 13:00 – 16:00 (Before Midterm Examination)

or Tuesday at 9:00 – 12:00 (After Midterm Examination)

or as an advance appointment

Outline

A decorative header image showing a close-up of a document with the Laplace Transform formula. The text "Laplace-Trans" is visible at the top, and the formula $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$ is written below it.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- Course Introduction
 - Syllabus
 - Calendar
- Linear Algebra
- Matrices
 - Definition of a Matrix
 - Matrix Addition
 - Scalar Multiplication
 - Matrix Multiplication
 - The Transpose of a Matrix

Course Introduction

A decorative image in the top right corner showing a close-up of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$. The text "Laplace-Trans" is also visible above the formula.
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- Course: 06016401
- Course Title: Mathematics for Information Technology
- Duration: 17 weeks (including Midterm and Final examination weeks)
- Credit Hours: 3 (3-0-6) credit hours
 - Lecture: 45 hours
 - Self-Study: 90 hours
- Lecturers
 - Asst. Prof. Dr. Somkiat Wangsiripitak
 - Asst. Prof. Dr. Praphan Pavarangkoon

Course Introduction (cont.)

- Teaching Assistants
 - 65070099 ธนาวัลย์ แซ่มเสถียร (มาร์ค)
 - Tuesday: 2 hours at 16:00 - 18:00
 - 65070174 ภาคิน จันทร์จำลอง (อึก)
 - Thursday: 2 hours at 16:00 - 18:00
 - 65070239 สุขานันท์ อุ่ณหมันกิจ (น้ำขิง)
 - Wednesday: 2 hours at 16:00 - 18:00
- Course Date/Time/Room
 - This class meets every Tuesday and Wednesday (3 hours/week).
 - Tuesday: 3 hours at 9:00 - 12:00 and 13:00 - 16:00
 - Wednesday: 3 hours at 13:00 - 16:00

Course Description

A decorative image in the top right corner showing a portion of a document with the Laplace Transform formula:
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

- ลิมิตและความต่อเนื่อง อนุพันธ์ของฟังก์ชัน อนุพันธ์อันดับสูง การประยุกต์ของอนุพันธ์ ปัญหาค่าสูงสุดและต่ำสุดของฟังก์ชัน การอินทิเกรตและเทคนิคของการอินทิเกรต การประยุกต์ของการอินทิเกรต ฟังก์ชันสองตัวแปร อนุพันธ์ย่อย พีชคณิตเชิงเส้น เมทริกซ์และดีเทอร์มิแนนต์ ระบบสมการเชิงเส้น และการหาผลเฉลยของระบบสมการเชิงเส้น เวกเตอร์สเปซ การแปลงเชิงเส้น การแปลงเมทริกซ์ เมทริกซ์เชิงตั้งฉาก

Aims and Objectives

The image shows a close-up of a document with the Laplace Transform formula. The text "Laplace-Trans" is visible at the top, and the formula $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$ is written below it.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- จุดมุ่งหมาย
 - เพื่อให้ผู้เรียนมีความรู้พื้นฐานทางคณิตศาสตร์ที่จำเป็นสำหรับการเรียนในวิชาต่างๆ เกี่ยวกับเทคโนโลยีสารสนเทศ
- วัตถุประสงค์
 - มีการประยุกต์ใช้ซอฟต์แวร์ที่เหมาะสมนอกเหนือจากการคำนวณแบบธรรมดา เพื่อให้ผู้เรียนสามารถประยุกต์ความรู้ที่ได้เรียนในการแก้ปัญหาทางเทคโนโลยีสารสนเทศได้

Topics and Details of 17 Weeks

Week	Date	Topics and Details	Lecturer
1	04/07 and 05/07	Linear algebra	Praphan
2	11/07 and 12/07	Matrices and determinants	Praphan
3	18/07 and 19/07	System of linear equations and its solutions	Praphan
4	01/08 and 02/08 (Online)	Vector space (1)	Praphan
5	08/08 and 09/08	Vector space (2)	Praphan
6	15/08 and 16/08	Linear transformations	Praphan
7	22/08 and 23/08	Orthogonal matrix	Praphan
8	Midterm examination Tuesday 29 August at 9:30 – 12:30		

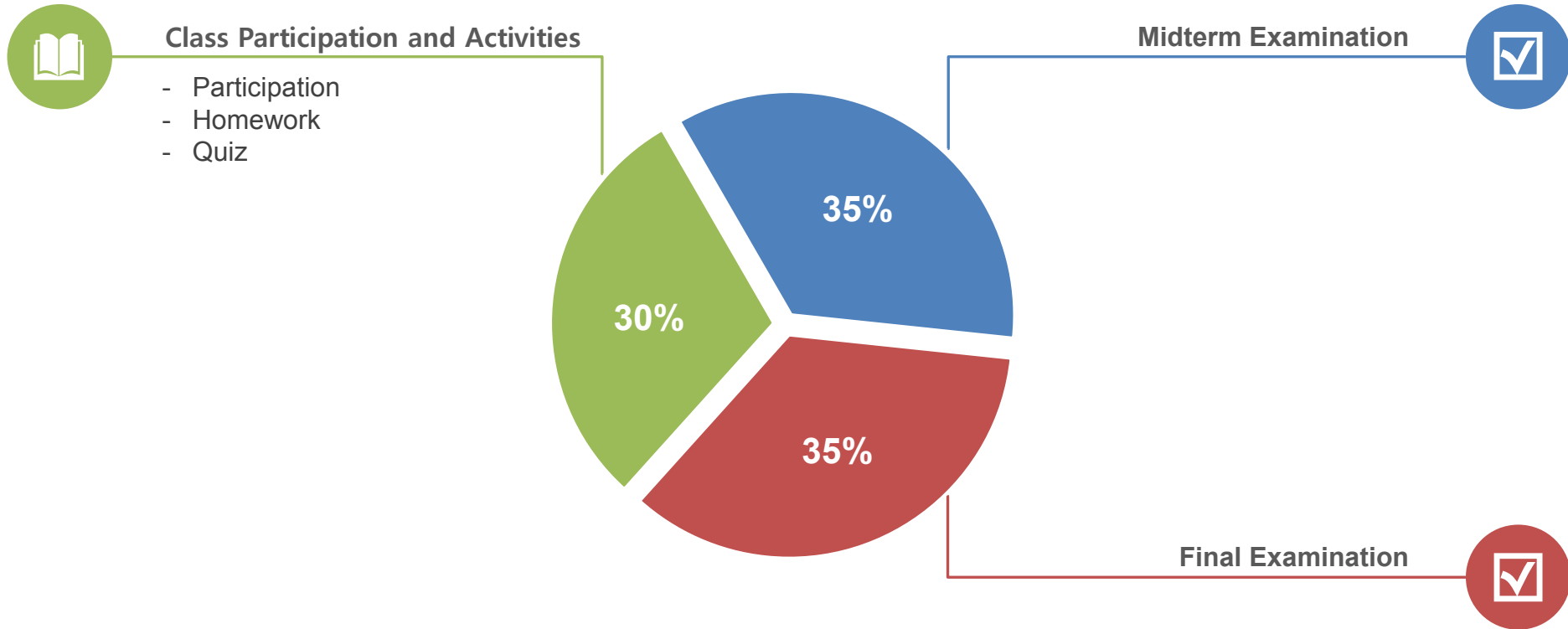
Topics and Details of 17 Weeks (cont.)

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Week	Date	Topics and Details	Lecturer
9	05/09 and 06/09	Function, limit and continuity	Somkiat
10	12/09 and 13/09	Derivative of functions	Somkiat
11	19/09 and 20/09	Higher-order derivatives, chain rule, implicit differentiation	Somkiat
12	26/09 and 27/09	Applications of the derivative	Somkiat
13	03/10 and 04/10	Integration (indefinite & definite)	Somkiat
14	10/10 and 11/10	Integration by substitution	Somkiat
15	17/10 and 18/10	Applications of integration	Somkiat
16	24/10 and 25/10	Function of two variables, partial derivatives	Somkiat
17	Final examination Tuesday 31 October at 9:30 – 12:30		

Course Evaluation and Evaluation Criteria

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$



Course Assessment

Laplace-Trans
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Description	Grade		Score (points)
Excellent	A	4.0	80-100
Very Good	B+	3.5	75-79
Good	B	3.0	70-74
Fairly Good	C+	2.5	65-69
Fair	C	2.0	60-64
Poor	D+	1.5	55-59
Very Poor	D	1.0	50-54
Fail	F	0.0	0-49

หมายเหตุ ผู้สอนอาจจะพิจารณาแบบอิงกลุ่มร่วมด้วย

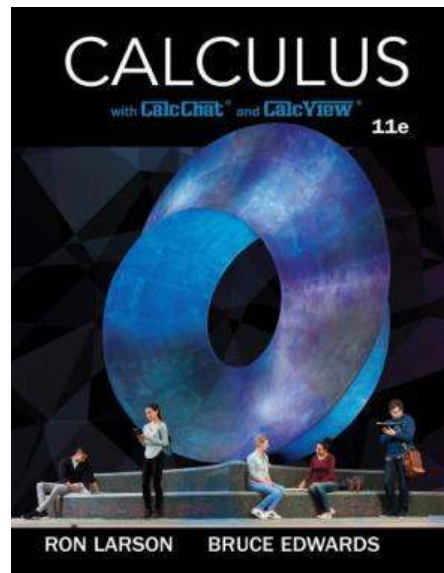
Resources

- ตำราและเอกสารหลัก

- Title: Calculus

Authors: Larson Edwards and Bruce H. Edwards

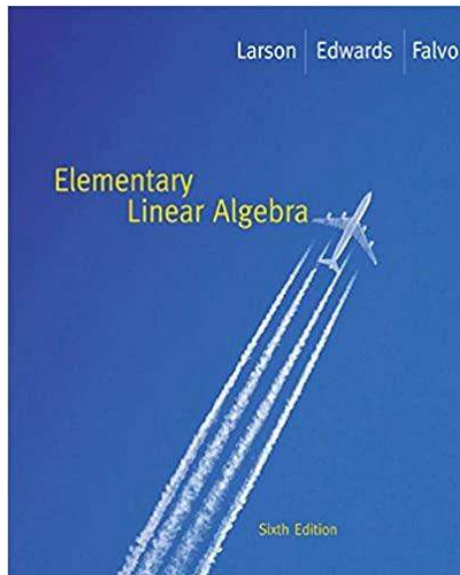
Publisher: Brooks/Cole, Cengage Learning



Resources (cont.)

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- Title: Elementary Linear Algebra
Authors: Ron Larson and David C. Falvo
Publisher: Houghton Mifflin Harcourt



Note

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- Student Responsibility, Course Policies (Attendance, Late/Absence/Medical Reasons), Disabilities and Accommodations, Academic Dishonesty (Plagiarism), and KMITL rules and regulations: Please refer to **Curriculum Guide**

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- $\rho_1 \cdot \frac{1}{2} \rho_2 \cdot \frac{1}{2} \rho_3 \cdot \frac{1}{2} \rho_4 \cdot \frac{1}{2} \rho_5 \cdot \frac{1}{2} \rho_6 \cdot \frac{1}{2} \rho_7 \cdot \frac{1}{2} \rho_8 \cdot \frac{1}{2} \rho_9 \cdot \frac{1}{2} \rho_{10}$

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๑๑ ค.บ.บัวภาคนันท์ อึ้ง, ค.บ.บัวสน

What Is Linear Algebra? (cont.)

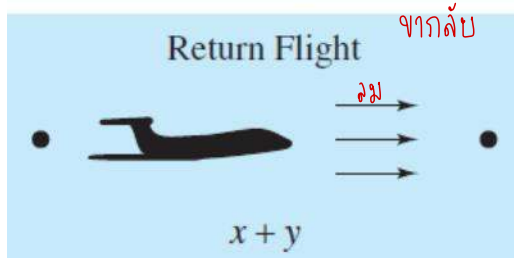
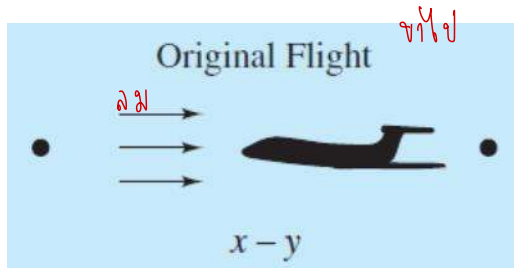
Laplace-Trans

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

x = ค.เร็วเครื่องบิน

y = ค.เร็วลม

ทิศทางเดียวกัน ค.เร็วเพิ่มขึ้นเลขเป็น +



- If you let x represent the speed of the plane and y the speed of the wind, then the following system models the problem.

เวลา (ค.เร็ว) = ระยะทาง
 6.25($x - y$) = 5000 $\Leftrightarrow t(v) = 5$

ทางกลับ $5(x + y) = 5000$ $\rightarrow x - y$ เพราะทิศทางตรงกันข้าม

- This system of two equations and two unknowns **simplifies** to

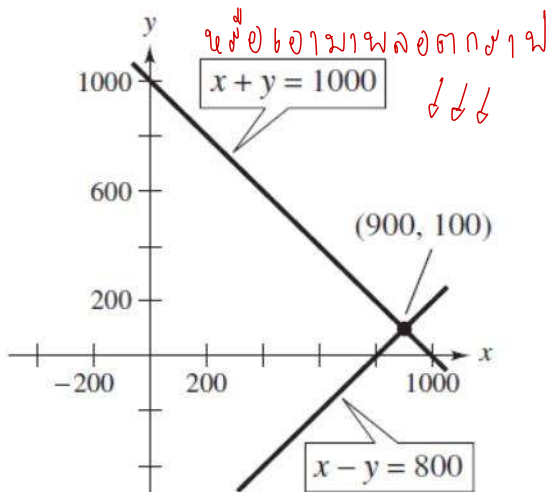
$x - y = 800$

จับ 2 สมการตั้งกันบวก, - กัน

$x + y = 1000$

หาค่าบวกแล้วแทนค่า

- and the solution is $x = 900$ kilometers per hour and $y = 100$ kilometers per hour.
- Geometrically, this system represents two lines in the xy -plane.
- You can see in the figure that these lines intersect at the point $(900, 100)$ which verifies the answer that was obtained.



The lines intersect at (900, 100).

What Is Linear Algebra? (cont.)

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$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- Solving systems of linear equations is one of the most important applications of linear algebra.
- It has been argued that the majority of all mathematical problems encountered in scientific and industrial applications involve solving a linear system at some point.
- Linear applications arise in such diverse areas as engineering, chemistry, economics, business, ecology, biology, and psychology.

Linear Algebra

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

ระบบสมการเชิงเส้น

- **Linear algebra** is a fairly extensive subject that covers vectors and matrices, determinants, systems of linear equations, vector spaces and linear transformations, eigenvalue problems, and other topics.
ปริภูมิเวกเตอร์
ก. ระบบสมการเชิงเส้น ค. ค่าเฉพาะ
- As an area of study it has a broad appeal in that it has many applications in engineering, physics, geometry, computer science, economics, and other areas.
- It also contributes to a deeper understanding of mathematics itself.

Definition of a Matrix

นิยามของเมทริกซ์

- A **matrix** is a rectangular array of numbers or functions which we will enclose in brackets.
- The plural of matrix is **matrices**.

If m and n are positive integers, then an $m \times n$ **matrix** is a rectangular array

element
สมาชิก

$$\left[\begin{array}{ccccc} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{array} \right]$$

n columns คอลัมน์แนวตั้ง!

m rows แถวแนวนอน!

in which each **entry**, a_{ij} , of the matrix is a number. An $m \times n$ matrix (read “ m by n ”) has m **rows** (horizontal lines) and n **columns** (vertical lines).

Examples of Matrices



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Each matrix has the indicated size.

(a) Size: 1×1

$$[2]$$

1 R 1 C

(b) Size: 2×2

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

2 R 2 C

(c) Size: 1×4

$$\left[1 \quad -3 \quad 0 \quad \frac{1}{2} \right]$$

1 R 4 C

(d) Size: 3×2

$$\begin{bmatrix} e & \pi \\ 2 & \sqrt{2} \\ -7 & 4 \end{bmatrix}$$

3 R 2 C

Definition of a Matrix (cont.)

- One very common use of matrices is to represent systems of linear equations.
- The matrix derived from the **coefficients** and **constant** terms of a system of linear equations is called the **augmented matrix** of the system.
- The matrix containing only the coefficients of the system is called the **coefficient matrix** of the system. Here is an example.

coefficients

System

$$\begin{array}{rclcl} x - 4y + 3z & = & 5 & ① \\ -x + 3y - z & = & -3 & ② \\ 2x & - & 4z & = & 6 \end{array} \quad \text{constant}$$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & -4 & 3 & 5 \\ -1 & 3 & -1 & -3 \\ 2 & 0 & -4 & 6 \end{array} \right] \quad \text{constant}$$

Coefficient Matrix

$$\left[\begin{array}{ccc} 1 & -4 & 3 \\ -1 & 3 & -1 \\ 2 & 0 & -4 \end{array} \right]$$

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Operations with Matrices

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- Matrix Addition การบวกเมทริกซ์
- Scalar Multiplication การคูณด้วยสเกลาร์
- Matrix Multiplication การคูณเมทริกซ์

Matrix Addition

ขนาดของเมทริกซ์ต้องเท่ากัน

- You can **add** two matrices (of the same size) by adding their corresponding entries.

ใช้อักษรพิมพ์ใหญ่เพื่อบ่งบอกว่า เป็นเมทริกซ์

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are matrices of size $m \times n$, then their sum is the $m \times n$ matrix given by $i = \text{rows}, j = \text{column}$

$$A + B = [a_{ij} + b_{ij}]$$

นำสมาชิกในตำแหน่งเดียวกันมา + กัน

The sum of two matrices of different sizes is undefined.

ขนาดต่างกัน = ไม่ได้

Addition of Matrices

Laplace-Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

(a) $\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -1+1 & 2+3 \\ 0-1 & 1+2 \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ -1 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(d) The sum of

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 0 & -1 \\ 3 & -2 & 2 \end{bmatrix}_{3 \times 3} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 \\ -1 & 3 \\ 2 & 4 \end{bmatrix}_{3 \times 2}$$

is undefined.

ขนาดไม่เท่ากัน

Scalar Multiplication

ก. คูณด้วยสเกลาร์

สเกลาร์ = จน.จริงใด ๆ

- When working with matrices, **real numbers are referred to as scalars.**
- You can multiply a matrix A by a scalar c by multiplying each entry in A by c .

If $A = [a_{ij}]$ is an $m \times n$ matrix and c is a scalar, then the **scalar multiple** of A by c is the $m \times n$ matrix given by

$$cA = [ca_{ij}]$$

๖

๖ $\text{cA} = [\text{c}a_{ij}]$ ๖ c scalar $\times$ สมาชิกทุกตัวในเมทริกซ์

Scalar $c \times$ matrix A

You can use $-A$ to represent the scalar product $(-1)A$. If A and B are of the same size, $A - B$ represents the sum of A and $(-1)B$. That is,

$$A - B = A + (-1)B.$$

Subtraction of matrices

Scalar Multiplication and Matrix Subtraction

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

For the matrices

$$A = \begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix}_{3 \times 3} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix}_{3 \times 3}$$

find (a) $3A$, (b) $-B$, and (c) $3A - B$.

SOLUTION

$$(a) \quad 3A = 3 \begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 3(1) & 3(2) & 3(4) \\ 3(-3) & 3(0) & 3(-1) \\ 3(2) & 3(1) & 3(2) \end{bmatrix} = \begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix}$$

$$(b) \quad -B = (-1) \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ -1 & 4 & -3 \\ 1 & -3 & -2 \end{bmatrix}$$

$$(c) \quad 3A - B = \begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 12 \\ -10 & 4 & -6 \\ 7 & 0 & 4 \end{bmatrix}$$

Matrix Multiplication



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

If $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{ij}]$ is an $n \times p$ matrix, then the **product** AB is an $m \times p$ matrix

$$AB = [c_{ij}]$$

where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj}.$$

- This definition means that the entry in the i th row and the j th column of the product AB is obtained by multiplying the entries in the i th row of A by the corresponding entries in the j th column of B and then adding the results.

Finding the Product of Two Matrices

$$\text{Laplace-Trans} \\ F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Find the product AB , where

$$A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}.$$

SOLUTION

First note that the product AB is defined because A has size 3×2 and B has size 2×2 . Moreover, the product AB has size 3×2 and will take the form

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}.$$

To find c_{11} (the entry in the first row and first column of the product), multiply corresponding entries in the first row of A and the first column of B . That is,

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 \\ -4 \end{bmatrix} = \begin{bmatrix} -9 & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}.$$

$c_{11} = (-1)(-3) + (3)(-4) = -9$

Finding the Product of Two Matrices (cont.)



Laplace Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

Similarly, to find c_{12} , multiply corresponding entries in the first row of A and the second column of B to obtain

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}.$$

$c_{12} = (-1)(2) + (3)(1) = 1$

Continuing this pattern produces the results shown below.

$$c_{21} = (4)(-3) + (-2)(-4) = -4$$

$$c_{22} = (4)(2) + (-2)(1) = 6$$

$$c_{31} = (5)(-3) + (0)(-4) = -15$$

$$c_{32} = (5)(2) + (0)(1) = 10$$

The product is

$$AB = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{bmatrix}.$$

Matrix Multiplication (cont.)

- Be sure you understand that for the product of two matrices to be defined, the number of columns of the first matrix must equal the number of rows of the second matrix. That is,

$$\begin{array}{ccc} A & B & = AB. \\ m \times n & n \times p & m \times p \\ \uparrow \quad \uparrow \quad \uparrow & & \\ \text{equal} & & \\ \text{size of } AB & & \end{array}$$

- So, the product BA is not defined for matrices such as A and B in the previous example.

Matrix Multiplication (cont.)

- The general pattern for matrix multiplication is as follows. To obtain the element in the i th row and the j th column of the product AB , use the i th row of A and the j th column of B .

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2p} \\ b_{31} & b_{32} & \cdots & b_{3j} & \cdots & b_{3p} \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nj} & \cdots & b_{np} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1j} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2j} & \cdots & c_{2p} \\ \vdots & \vdots & & \vdots & & \vdots \\ c_{i1} & c_{i2} & \cdots & c_{ij} & \cdots & c_{ip} \\ \vdots & \vdots & & \vdots & & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mj} & \cdots & c_{mp} \end{bmatrix}$$

$a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj} = c_{ij}$

Matrix Multiplication



Laplace-Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

$$(a) \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -2 \end{bmatrix} \begin{bmatrix} -2 & 4 & 2 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} -5 & 7 & -1 \\ -3 & 6 & 6 \end{bmatrix}$$

$2 \times 3 \quad 3 \times 3 \quad 2 \times 3$

$$(b) \begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix}$$

$2 \times 2 \quad 2 \times 2 \quad 2 \times 2$

$$(c) \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$2 \times 2 \quad 2 \times 2 \quad 2 \times 2$

$$(d) \begin{bmatrix} 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix}$$

$1 \times 3 \quad 3 \times 1 \quad 1 \times 1$

$$(e) \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -4 & -6 \\ -1 & 2 & 3 \\ 1 & -2 & -3 \end{bmatrix}$$

$3 \times 1 \quad 1 \times 3 \quad 3 \times 3$

Properties of Matrix Operations

- This section begins to develop the **algebra of matrices**.
- You will see that this algebra shares many (but not all) of the properties of the algebra of real numbers.

Properties of Matrix Addition and Scalar Multiplication

If A , B , and C are $m \times n$ matrices and c and d are scalars, then the following properties are true.

1. $A + B = B + A$
2. $A + (B + C) = (A + B) + C$
3. $(cd)A = c(dA)$
4. $1A = A$
5. $c(A + B) = cA + cB$
6. $(c + d)A = cA + dA$

Commutative property of addition

Associative property of addition

Associative property of multiplication

Multiplicative identity

Distributive property

Distributive property

Properties of Matrix Multiplication

If A , B , and C are matrices (with sizes such that the given matrix products are defined) and c is a scalar, then the following properties are true.

1. $A(BC) = (AB)C$
2. $A(B + C) = AB + AC$
3. $(A + B)C = AC + BC$
4. $c(AB) = (cA)B = A(cB)$

Properties of the Identity Matrix



Laplace Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

- You will now look at a special type of square matrix that has 1's on the main diagonal and 0's elsewhere.

$$I_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

For instance, if $n = 1, 2$, or 3 , we have

$$I_1 = [1], \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

1×1 2×2 3×3

- When the order of the matrix is understood to be n , you may denote I_n simply as I .

Properties of the Identity Matrix (cont.)

A decorative image in the top right corner showing a close-up of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$. The text "Laplace-Trans" is also visible above the formula.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

If A is a matrix of size $m \times n$, then the following properties are true.

1. $AI_n = A$
2. $I_m A = A$

As a special case of this theorem, note that if A is a *square* matrix of order n , then

$$AI_n = I_n A = A.$$

The Transpose of a Matrix

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

The **transpose** of a matrix is formed by writing its columns as rows. For instance, if A is the $m \times n$ matrix shown by

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix},$$

Size: $m \times n$

then the transpose, denoted by A^T , is the $n \times m$ matrix below

$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} & \cdots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \cdots & a_{m2} \\ a_{13} & a_{23} & a_{33} & \cdots & a_{m3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{mn} \end{bmatrix}.$$

Size: $n \times m$

The Transpose of a Matrix (cont.)

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the transpose of each matrix.

$$(a) A = \begin{bmatrix} 2 \\ 8 \end{bmatrix} \quad (b) B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad (c) C = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (d) D = \begin{bmatrix} 0 & 1 \\ 2 & 4 \\ 1 & -1 \end{bmatrix}$$

SOLUTION

$$(a) A^T = \begin{bmatrix} 2 & 8 \end{bmatrix} \quad (b) B^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \quad (c) C^T = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(d) D^T = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & -1 \end{bmatrix}$$

- Note that the square matrix in part (c) of the above example is equal to its transpose.
- Such a matrix is called **symmetric**.
- A matrix A is symmetric if $A = A^T$. From this definition it is clear that a symmetric matrix must be square. Also, if $A = [a_{ij}]$ is a symmetric matrix, then $a_{ij} = a_{ji}$ for all $i \neq j$.

Properties of Transposes

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

If A and B are matrices (with sizes such that the given matrix operations are defined) and c is a scalar, then the following properties are true.

1. $(A^T)^T = A$ Transpose of a transpose
2. $(A + B)^T = A^T + B^T$ Transpose of a sum
3. $(cA)^T = c(A^T)$ Transpose of a scalar multiple
4. $(AB)^T = B^T A^T$ Transpose of a product

The Product of a Matrix and Its Transpose

For the matrix

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -2 & -1 \end{bmatrix}$$

find the product AA^T and show that it is symmetric.

SOLUTION

Because

$$AA^T = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 3 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 10 & -6 & -5 \\ -6 & 4 & 2 \\ -5 & 2 & 5 \end{bmatrix}$$

it follows that $AA^T = (AA^T)^T$, so AA^T is symmetric.

Q & A

A close-up, slightly blurred image of a document showing the Laplace transform formula. The text "Laplace-Trans" is visible at the top, followed by the equation $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$ and some faint text below it.

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$
with initial...